

SALE, Morocco

WMO: 601350

Lat: 34.051N

Lon: 6.752W

Elev: 84

StdP: 100.32

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.8	(c) 5.9	(d) 1.0	(e) 4.1	(f) 11.8	(g) 2.8	(h) 4.7	(i) 10.0	(j) 10.3	(k) 14.6	(l) 8.6	(m) 14.9	(n) 1.3	(o) 140	(p) 0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.5	(c) 32.8	(d) 21.0	(e) 30.0	(f) 21.0	(g) 28.1	(h) 21.1	(i) 24.0	(j) 28.1	(k) 23.1	(l) 26.9	(m) 22.4	(n) 26.0	(o) 4.2	(p) 340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	17.8	26.5	22.0	16.8	25.3	21.2	16.0	24.4	72.5	28.2	68.9	27.1	66.3	26.0	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.5	7.3	6.4	2.1	39.5	1.8	3.5	0.7	42.0	-0.3	44.1	-1.4	46.0	-2.7	48.5
			1.3	25.2	1.8	2.1	0.0	26.7	-1.1	27.9	-2.1	29.0	-3.5	30.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.7	12.4	12.9	14.9	16.0	18.4	20.8	22.4	22.9	21.6	19.7	16.0	13.7	(6)
	DBStd	4.33	2.10	2.23	2.57	2.26	2.65	2.51	2.29	2.01	1.96	2.31	2.60	2.19	(7)
	HDD10.0	7	3	3	0	0	0	0	0	0	0	0	0	1	(8)
	HDD18.3	795	183	152	114	77	30	4	0	0	1	9	81	146	(9)
	CDD10.0	2809	79	84	151	181	259	324	385	402	349	302	179	114	(10)
	CDD18.3	554	0	1	6	7	31	78	127	143	99	53	9	1	(11)
	CDH23.3	3271	2	6	68	83	249	433	739	872	492	274	49	3	(12)
(13) Wind	CDH26.7	833	0	1	21	27	101	138	193	182	88	74	8	0	(13)
	WSAvg	2.8	2.5	2.7	3.0	3.2	3.2	3.1	3.0	2.8	2.7	2.6	2.6	2.7	(14)
	PrecAvg	563	79	63	58	57	19	6	0	1	8	48	88	116	(15)
	PrecMax	866	344	202	228	194	96	31	10	9	62	166	343	394	(16)
	PrecMin	286	0	0	0	1	0	0	0	0	0	0	0	0	(17)
	PrecStd	160	70	50	49	50	20	8	2	2	13	46	78	88	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.1	24.1	29.6	30.8	34.8	36.2	37.0	35.2	32.4	33.2	27.8	23.1	(19)
		MCWB	15.7	14.9	18.3	18.8	19.8	21.6	24.3	21.4	22.4	18.2	18.4	15.5	(20)
	2%	DB	19.3	20.9	24.8	25.2	29.2	30.0	30.5	30.3	28.9	28.2	24.4	20.9	(21)
		MCWB	14.7	15.6	16.4	17.1	19.2	20.6	22.2	22.8	21.3	18.5	17.3	15.2	(22)
	5%	DB	18.0	18.9	21.8	22.4	25.5	27.2	28.1	28.2	27.0	25.9	22.3	19.2	(23)
		MCWB	14.5	15.0	16.0	16.9	18.6	20.6	22.4	22.4	21.1	19.3	17.5	15.2	(24)
	10%	DB	16.9	17.6	19.8	20.9	23.2	25.1	26.6	27.0	25.8	24.2	20.9	18.1	(25)
		MCWB	13.8	14.4	15.6	16.4	18.3	20.2	21.8	22.1	20.7	19.2	16.9	14.8	(26)
	0.4%	WB	17.3	17.8	19.4	20.1	22.0	23.7	25.8	25.6	24.3	22.3	20.6	18.1	(27)
		MCDB	19.8	21.1	25.3	26.2	27.6	30.1	31.6	29.9	28.8	25.9	23.7	20.3	(28)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	15.8	16.5	17.8	18.5	20.3	22.1	24.0	24.0	22.8	21.2	19.4	16.9	(29)
		MCDB	18.2	19.2	22.3	23.1	25.6	26.8	28.3	27.4	26.3	24.4	22.3	19.1	(30)
	5%	WB	15.0	15.5	16.9	17.5	19.5	21.2	22.8	23.2	22.1	20.6	18.4	16.0	(31)
		MCDB	17.4	18.0	20.4	21.5	24.1	25.4	26.7	26.6	25.3	23.6	21.4	18.4	(32)
	10%	WB	14.1	14.8	16.0	16.8	18.6	20.6	22.1	22.5	21.4	19.9	17.3	15.2	(33)
		MCDB	16.4	17.1	19.1	20.2	22.7	24.5	25.7	25.9	24.6	23.0	20.2	17.7	(34)
	5% DB	MDBR	9.2	9.5	10.0	9.6	9.5	8.9	8.4	8.5	8.9	9.6	9.5	9.2	(35)
		MCDBR	10.6	11.4	14.1	13.3	14.7	12.9	11.4	10.8	10.7	12.2	11.2	10.5	(36)
		MCWBR	7.0	7.3	7.9	7.4	6.9	5.7	5.4	5.1	5.3	6.0	6.4	6.6	(37)
		MCWBR	9.2	10.0	11.3	11.2	11.2	10.0	9.8	9.0	9.1	9.6	9.5	8.5	(38)
(38) Clear-Sky Solar Irradiance	taub		0.330	0.357	0.409	0.405	0.429	0.461	0.498	0.493	0.441	0.403	0.363	0.338	(40)
		taud	2.509	2.420	2.251	2.293	2.249	2.157	2.033	2.065	2.244	2.362	2.455	2.505	(41)
	Ebn at Noon		881	888	863	883	861	830	796	793	821	826	832	848	(42)
		Edh at Noon	84	102	131	131	139	152	171	163	130	107	89	81	(43)
(44) All-Sky Solar Radiation	RadAvg		2.94	3.83	5.01	6.12	6.94	7.62	7.66	6.92	5.65	4.18	3.12	2.67	(44)
	RadStd		0.24	0.36	0.36	0.31	0.41	0.33	0.17	0.18	0.23	0.31	0.24	0.19	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	-1.11	N/A	-0.74	-0.79	N/A	N/A	N/A	N/A
Regional (5 neighbors)		N/A	N/A	N/A	+1.01	N/A	N/A	N/A	N/A	N/A	+85

Nomenclature: See separate page